

Received 20 May 2025, accepted 4 June 2025, date of publication 16 June 2025, date of current version 8 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3579755

RESEARCH ARTICLE

Efficient Machine Learning Models for Solar Radiation Prediction Using Ensemble Techniques: A Case Study in Low-Rainfall Arid Climates

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ABSTRACT Solar energy is a green energy source that can be used to generate electricity, making it one of the oldest existing renewable energy sources. However, solar energy values are not always available in many areas far from urban cities, due to the high cost of purchasing and maintaining measuring instruments. In these areas, the only way to know the values of solar radiation from is to predict these values. To address this problem, we have designed an efficient model capable of forecasting hourly solar radiation a climate with a combination of arid and humid conditions using real-time data from meteorological variables measured by a Meteorological Station located in a university area isolated from the central city. For the development of our model, 11 meteorological variables were collected from January to December 2023. From this, dimensionality reduction was carried out using the Principal Component Analysis (PCA) technique, obtaining the 8 most appropriate representative variables for the prediction of solar radiation using different Machine Learning (ML) models. To ensure the reliability of our model, we compared the performance metrics of several ML models. Finally, ensemble techniques were applied to enhance prediction accuracy by combining various ML algorithms. The best results were obtained using the Stacking Regressor (SR), achieving an R^2 of 0.92, RMSE of 47.30 W/m², and MAE of 18.27 W/m², demonstrating high predictive performance in the target climate region.

INDEX TERMS Machine learning, meteorology, prediction, solar radiation.

I. INTRODUCTION

The energy released by the Sun, transmitted through electromagnetic radiation, is known as solar radiation. This energy is a mixture of radiation with wavelengths between 200 and 4000 nm, which are called ultraviolet radiation, visible light, and infrared radiation. In turn, the magnitude that measures the solar radiation that reaches the Earth is the irradiance, also called global solar radiation, which is defined as the power per unit of area, and is measured in (W/m²).

Solar energy is known to be a renewable green energy source and this energy can be harnessed with the use of

photovoltaic techniques to generate electricity. According to [1], it is mentioned that solar radiation data is necessary in various applications such as solar panel, agricultural crop models, air quality models, and hydrological models.

On the other hand, in [2], it is mentioned that the continuous use of fossil energy has caused a serious environmental pollution problem; therefore, to solve this serious problem, the use of renewable energy is proposed. A source of renewable energy is solar energy that can generate the demand for electricity necessary for industries and the population in general. In [3] they mentioned that renewable energies, such as solar and wind, have been gaining popularity for more than a decade and could be used in rural, underdeveloped areas that desire energy independence.

The associate editor coordinating the review of this manuscript and approving it for publication was Kai Song¹.

However, changes in solar radiation over time are caused by meteorological conditions, meaning factors like cloud cover and air pollution can influence the intensity of solar radiation [2].

In the study by [4], it is stated that predicting global solar radiation in a particular area is crucial, as it offers investors valuable information on the solar resources of that region, which can be highly advantageous for the development of large-scale solar energy projects. According to [5], solar radiation is an essential parameter for the design and efficient operation of solar energy systems. In many locations, records of solar radiation are not accessible because of the high costs associated with acquiring and maintaining the necessary measurement instruments. As a result, it is important to predict solar radiation with a reasonable degree of accuracy for these areas using different solar radiation estimation models.

A classic form of prediction from historical data commonly used is the use of linear or polynomial regression models. However, in our work we implement a Solar Radiation prediction model using Machine Learning (ML) algorithms based on the most representative values of the meteorological data obtained during the year 2023. This is because ML based methods outperform classical regression methods for predictive modeling due to several reasons such as:

- They handle nonlinearities in the data more effectively than classical regression methods, which assume linear relationships.
- They identify patterns between variables that may not be evident with classical regression methods, leading to more accurate predictions.
- They adapt to the presence of outliers, missing values, or measurement errors.
- They perform an automatic selection of features, identifying the most relevant features for prediction.
- They are able to handle large amounts of data and complex problems, making them suitable for large-scale predictive modeling.

Our study area is the National University of Engineering (UNI) and is located in the northern area of the city of Lima, where it is known that it is an area where solar energy greatly affects. Having difficulties in obtaining these values of incident energy due to the lack of measuring devices, it is essential to know these monthly values of solar energy, so that it can be used for other purposes such as the use of photovoltaic cells.

This challenge is particularly relevant in the context of Sustainable Development Goal 13 (SDG 13) established by the United Nations, which calls for urgent action to combat climate change and its impacts. Accurate solar radiation prediction models contribute to this goal by facilitating the deployment of renewable energy systems, especially in arid and under-resourced areas, thereby supporting climate adaptation and mitigation strategies [6].

Lima, Peru, has a climate classified as subtropical desert (according to the Köppen climate classification) [7], [8], [9].

This type of climate is characterized by a unique combination of arid and humid conditions, mainly influenced by its proximity to the Pacific Ocean, the cold Humboldt Current, and the proximity to the Andes. The main characteristics of this climate include:

- **Low Precipitation:** Lima is one of the driest capitals in the world, with annual rainfall generally not exceeding 10 mm, which significantly limits the development of vegetation.
- **High Humidity:** Despite the scarcity of rainfall, the city experiences high levels of humidity due to the moderating effect of the ocean and the Humboldt Current.
- **Mild Temperatures:** Lima's climate is moderate year-round, with cool winters (12–19 °C or 54–66 °F) and warm summers (12–29 °C or 64–84 °F).
- **Frequent Presence of Fog:** “Garúa,” a typical coastal fog, is common in the winter months and contributes to a cool, cloudy environment.

From a technical point of view, Lima's climate combines three main elements of the Köppen classification:

- 1) **Desert (B):** Characterized by extremely low annual precipitation, insufficient to support significant vegetation.
- 2) **Hot (W):** Represents arid regions with consistently high temperatures, especially during the warmer months.
- 3) **Subtropical (h):** Indicates a warm climate with an annual mean temperature above 18 °C (64.4 °F).

Together, these features make Lima's climate exceptional compared to other typical desert climates, lacking extreme heat variations and being marked by unique coastal conditions such as persistent fog.

To carry out our study, a medium-sized weather station was installed at the UNI facilities, with the aim of collecting various meteorological data. From this information, monthly predictions of solar radiation could be made. It should be noted that there are numerous studies worldwide that estimate solar radiation using different developed models. In our case, the model was specifically applied to a city characterized by an arid climate and low rainfall, which adds a particular context to the analysis.

As a result of this study, we developed two predictive models using the data collected throughout 2023. The first model utilized traditional Machine Learning methods, while the second employed advanced Ensemble techniques. These Ensemble methods combine the outputs of multiple base models to enhance overall performance, boost prediction accuracy, and create a more resilient model capable of handling the inherent variations and complexities present in meteorological data.

II. REVIEW OF LITERATURE

The study in [10] highlights the need for more accurate daily irradiation forecasts in Singapore. To address this, they developed a climate research and prediction model using multivariate statistical learning techniques. This model provided hourly daily irradiation forecast data for three

years. The results were compared with those from the Global Forecast System, a numerical weather prediction model used by the U.S. National Oceanic and Atmospheric Administration (NOAA).

In [11], a daily forecasting model for hourly solar irradiation was proposed, which does not require historical irradiation data but uses some meteorological data. The model employed a recurrent neural network (LSTM-RNN) to manage long- and short-term memory and was compared with a forward neural network (FFNN). Experiments were conducted using meteorological data from stations in various countries representing different climates. The results showed that the LSTM-RNN approach was more accurate than the FFNN. Then in [12] they carried out an estimate of solar radiation east of the Mediterranean in Turkey, taking data from the Turkish State Meteorological Service. The authors used an ANN model to perform this estimation. In their work, the best estimation performance was obtained by optimizing the number of neurons used in the hidden layer of the network with the trial-and-error method.

Independently, in [13], compared different prediction models for solar radiation, focusing on multi-layer perceptron (MLP), boosted decision trees, and a new combination of these models with linear regression for predicting daily global solar irradiation (DGSR). The models were validated using real data from southern Algeria. Results showed that the MLP model outperformed the other models. In [14], both an ANN model and a Recurrent Neural Network (RNN) model were developed for solar radiation prediction. The authors utilized meteorological data from a weather station in Alabama. Their results indicated that the RNN model achieved higher prediction accuracy compared to the ANN model. Then, in [15], the performance of various evolutionary neural network models was evaluated to address the solar radiation prediction problem in Toledo, Spain. The prediction relied on satellite measurements and variables, eliminating the need for data from ground stations or atmospheric soundings. The authors reported that the results obtained for solar radiation estimation at the Toledo radiometric station demonstrated excellent performance with the evolutionary neural networks tested.

Independently in [16] developed a model for monthly solar radiation prediction in Turkey using multi-gene genetic programming (MGGP), an evolutionary algorithm based on symbolic genetic programming. The study compared the performance of the MGGP approach against three classical empirical equations widely used in the literature. Using long-term meteorological data from 163 weather stations across seven climatic regions of the country, the authors demonstrated the superiority of the AI-based model in generating more accurate and practical equations.

In the work of [17], the importance of predicting global solar radiation for the development of solar energy conversion systems was emphasized. The objective was to predict solar radiation data for four provinces in Turkey using machine learning algorithms such as ANN, K-NN, SVM, and Deep Learning (DL). The models were trained with meteorological

data collected by the Turkish State Meteorological Service for the years 2018-2019. The study concluded that the ANN algorithm was the most accurate. Then, in [18] mentioned that several types of ANN were developed to predict solar radiation in Makurdi, Nigeria. In their work they used: A Forward Backpropagation Neural Network (FFNN), a Radial Basis Function Network (RBFN) and Generalized Regression Neural Network (GRNN). The objective was to improve prediction accuracy by using an ANN ensemble for the average monthly global solar radiation.

Independently, in the work of [5], it was built an ANN model to reliably estimate solar radiation. In their work the authors investigate different ANN models with three different algorithms. These models were trained with meteorological data collected over one year from six hot-climate locations in Tamil Nadu, India. In the work of [4] the prediction of global solar radiation in 27 European cities was carried out. In their work they used 6 different Machine Learning algorithms such as: Linear model (LM), Decision Tree (DT), Support Vector Machine (SVM), Deep Learning (DL), Random Forest (RF) and Gradient Boosted Trees (GBT). The study concluded that the Support Vector Machine (SVM) algorithm provided the most accurate predictions.

Then, in [19], two ANN models were utilized for predicting daily solar radiation based on meteorological data. The first model was a multilayer perceptron neural network (MLP-NN), while the second was a nonlinear autoregressive network with exogenous inputs (NARX). Independently, in [20], a new hybrid approach for global solar radiation (GSR) prediction was introduced, leveraging Deep Learning techniques. These predictive models are particularly beneficial for optimizing the performance of photovoltaic power systems in the solar energy industry. The system proposed by the authors was tested in a GSR daily forecast system of six solar parks in Australia.

In the work of [21], they presented a study comparing 2 empirical models against 3 machine learning models such as: Artificial Neural Networks (ANN), Support Vector Machine (SVM) and Long Short-Term Memory (LSTM). Using monthly data from 6 stations in Turkey, they found that in terms of its metrics, the best model was the LSTM, followed by the ANN and SVM. In their work, the authors concluded that it is more appropriate to use machine learning models than empirical equations. Then, in [22], they indicated that the estimation of monthly Solar Radiation was carried out using five different approaches based on machine learning. The models used are Support Vector Machine Regression (SVMR), Long Short-Term Memory (LSTM), Gaussian Process Regression (GPR), Extreme Learning Machines (ELM) and K Nearest Neighbors (KNN). In their work, the authors found that the LSTM and GPR algorithms are an alternative for forecasting Solar Radiation in Turkey, which has arid and semi-arid climatic regions.

In [23], the authors explored the use of single neural networks (SNN) and bootstrapped aggregate neural networks (BANN) to predict hourly global solar radiation. In their

research, the authors acquire data from the Bouzareah-Algeria Renewable Energy Development Center. The authors concluded that the BANN model is particularly suitable for predicting solar radiation in locations without measurement equipment, such as solarimeters or pyrometers.

In [24] developed DeepSI, a Bayesian deep learning model combining bidirectional LSTM autoencoders and transformers with Monte Carlo dropout for long-term solar irradiance forecasting using downscaled climate data. Unlike their long-term focus, this study applies deep learning techniques for real-time, short-term prediction based on observed meteorological data in a subtropical desert climate. Finally, [25] conducted a comparative study using real data from a solar plant in Turkey, applying machine learning models and time series techniques to predict irradiance and temperature at 5-minute intervals. Their work highlights models such as Random Forest and MLP, focusing on improving the performance estimation of photovoltaic systems.

A. BIBLIOMETRIC ANALYSIS OF THE RESEARCH FIELD

In order to complement the literature review and gain a clearer understanding of current research trends and the thematic structure within the domain of this study, a bibliometric approach was employed using the VOSviewer software. The objective was to visualize the co-occurrence of keywords and the evolution of thematic focus across recent literature.

The analysis was based on 98 documents retrieved from the Scopus database using the keywords “solar radiation,” “machine learning,” “prediction,” and “meteorology.” Although the search initially covered publications from 2015 to 2025, the resulting dataset consisted of articles published between 2021 and 2024. The results of these processes are shown below.

In Figure 1 the nodes represent keywords extracted from the analyzed articles; the size indicates their frequency, and the colors group the terms into thematic clusters. A strong relationship can be observed among prediction and machine learning terms (red), environmental variables (green), and agricultural or meteorological applications (blue).

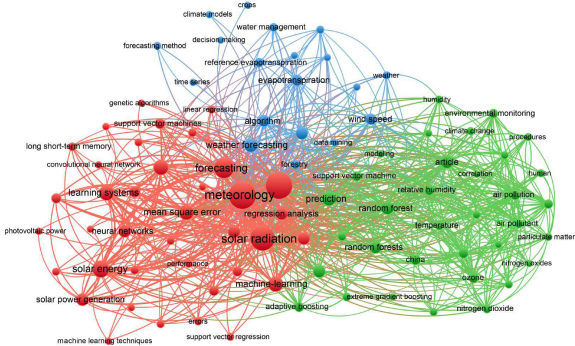


FIGURE 1. Keyword co-occurrence map generated with VOSviewer based on Scopus data.

In Figure 2 the color of each node reflects the average publication year of the corresponding keyword. Blue tones indicate earlier usage, while yellow tones represent more recent and emerging terms in the literature related to solar radiation, machine learning, prediction, and meteorology.

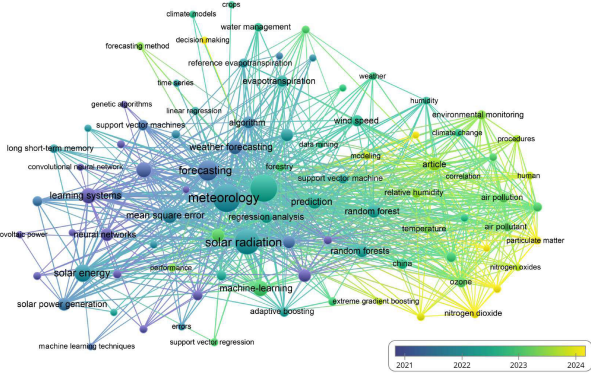


FIGURE 2. Overlay visualization of keyword co-occurrence generated with VOSviewer based on Scopus.

It should be noted that this bibliometric analysis was carried out independently of the article’s reference list and is intended solely to provide a broader contextual overview of the research domain based on Scopus-indexed publications.

III. MATERIALS AND METHODS

A. METHODOLOGY FOR THE DESIGN OF THE MACHINE LEARNING MODEL

The stages used in our study to achieve the contribution of our research are described below:

- Stage 1
The data are obtained from the meteorological station located at the University, which include various meteorological parameters recorded by the station.
- Stage 2
The next step in developing our model is to specify the input and output data. Initially, the values obtained from the weather station can be used as input data such as: Outside temperature, Wind chill, Dew point, Heat index, Pressure, Relative humidity, Rainfall, Wind direction, Wind speed, Average speed of wind, and Peak wind speed, while the value of solar radiation will be used in the output
- Stage 3
Involves preparing the data, analyzing null values, and performing data cleaning along with normalization.
- Stage 4
In this study, the data collected from the weather station is processed using statistical techniques, such as Principal Component Analysis (PCA), to identify the most representative variables for use as input data in our model. This approach helps to reduce the number of input variables.
- Stage 5

This step involves splitting the data of our model into separate sets for training and testing purposes.

- Stage 6
Apply various supervised machine learning regression models, including Random Forest (RF), Decision Trees (DT), Polynomial Regression (PR), Artificial Neural Networks (ANN), K-Nearest Neighbors (KNN) and Gradient Boosting (GB).
- Stage 7
At this stage we use Ensembles techniques, based on a combination of different ML algorithms with the aim of improving the precision of our results. To achieve them we use techniques such as: Voting, Stacking, Bagging and Boosting.
- Stage 8
Performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Nash-Sutcliffe Efficiency (NSE) and R^2 are calculated for each model during the training process. The models are then compared to determine which one achieves the best performance in predicting the output variable.

Figure 3 shows the stages.

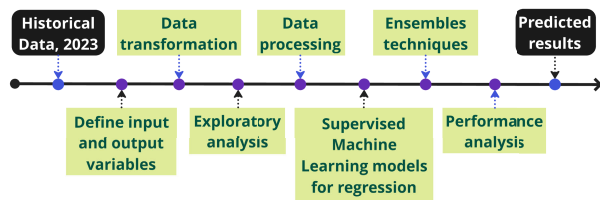


FIGURE 3. Stages to carry out our proposal.

For model analysis, we use Python 3.0 and a set of libraries used for machine learning and data processing, including Scikit-learn, Pandas, Matplotlib, Seaborn, TensorFlow, Keras and PyCaret. All testing was performed on a high-performance university laboratory workstation featuring an Intel Core i9 processor, 7 TB of storage, 128 GB of RAM, and an NVIDIA RTX 3090 GPU with 24 GB of VRAM.

B. DATA COLLECTION

In this stage, various parameters will be taken with a meteorological station acquired from Model: MKIII-RTI-LR, RAINWISE brand, installed on the roof of the Docimasia Laboratory of the Faculty of Geological, Mining and Metallurgical Engineering (FIGMM) of the UNI, as shown in Figure 4.

These parameters were taken throughout the day, from January to December 2023. It is known that Lima has a subtropical desert climate, a Meteorological Station was needed to determine the following parameters: Wind Speed, Wind Direction, Ambient Temperature, Relative Humidity, Barometric Pressure and Precipitation. These parameters were categorized into the following variables:

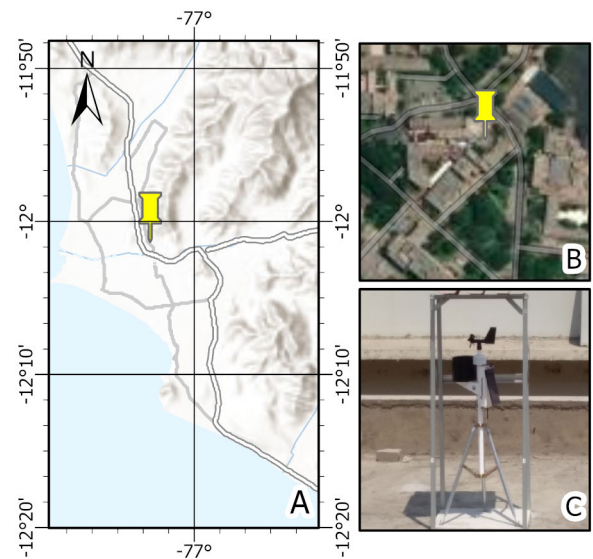


FIGURE 4. Location and view of the meteorological station installed at the National University of Engineering (UNI), Lima, Peru. (A) Regional map of Lima showing the station location. (B) Satellite image of the UNI campus, Docimacia area. (C) Photograph of the automated meteorological station installed on-site. Coordinates: -12.02246, -77.04803 (WGS84).

- Outside Temperature (OT): It is the temperature of the outside air.
- Wind Chill (Chill): Is an index that is used to evaluate the thermal sensation that a human body experiences under the combined effects of temperature and wind speed.
- Dew Point (Dew): It is the temperature to which the air must be cooled so that water vapor condenses into dew or frost.
- Heat Index (Heat): It is a measure that reflects what the human body perceives as temperature, factoring in both the ambient air temperature and the relative humidity.
- Relative Humidity (Humidity): It is the measure of the water vapor content in the air, expressed as a percentage.
- Pressure: Pressure.
- Rainfall: Rain.
- Wind Dir (Dir): Wind direction.
- Wind Speed (Wspeed): Wind speed.
- Avg. Wind Speed (Avwspeed): Average wind speed.
- Peak Wind Speed (Pwspeed): Maximum wind speed.

The input data obtained from our Meteorological Station a sample of the data is shown in Table 1.

Table 2 presents a statistical summary of the meteorological variables used in this study. Key statistics highlight the variability in certain variables, including Wind Direction, Peak Wind Speed, and Average Wind Speed, which show high coefficients of variation, suggesting considerable dispersion in the dataset. Additionally, variables such as Rainfall and Wind Speed exhibit significant skewness and/or high kurtosis, indicating skewed or fat-tailed distributions that could affect the accuracy of predictive models. Techniques such as normalization or transformation may be necessary to address these issues and improve model performance.

TABLE 1. Some data obtained from the meteorological station 22/02/2023.

Hour	OT	Chill	Dew	Heat	Hum	Pres	Rain	Dir	Ws	Avws	Pws
10:31	30.4	30.4	21.2	33.3	58	29.65	0	337	3	1	3
10:32	30.4	30.4	21.2	33.3	58	29.65	0	315	3	2	3
10:33	30.4	30.4	21.2	33.3	58	29.65	0	315	2	1	3
10:34	30.4	30.4	21.2	33.3	58	29.64	0	315	3	3	4
10:34	30.4	30.4	21.2	33.3	58	29.64	0	315	3	3	4
10:35	30.4	30.4	21.2	33.3	58	29.65	0	292	0	0	3
10:36	30.4	30.4	21.2	33.3	58	29.65	0	67	0	0	2

TABLE 2. Summary statistics of metereological variables.

Variable	Mean	Std	Min	Max	CV%	Skewness	Kurtosis
OT	21.687	4.150	0.1	33.3	22.351	-1.771	7.276
Chill	20.323	4.345	0.16	33.3	24.386	-0.286	1.077
Dew	16.628	8.648	0.1	83.1	15.503	0.394	4.345
Heat	20.620	5.251	0.1	39.4	29.376	0.196	0.304
Hum	77.524	12.623	1.0	96.0	20.240	-2.681	11.009
Press	27.294	9.683	0.008	92.0	38.533	-1.837	1.850
Rain	0.043	1.168	0.0004	80.0	49.989	0.011	0.002
Dir	195.977	128.207	1.0	360.0	61.282	-0.654	-1.256
WS	0.385	0.773	0.1667	9.0	144.589	0.962	-0.585
Avws	0.880	1.428	1.0	189.0	285.921	1.782	1.740
Pws	5.340	30.999	1.0	604.0	178.676	3.456	12.478

1) TREATMENT OF THE DATA OBTAINED

Nowadays, most scientific disciplines generate large volumes of data from their experiments. However, a significant portion of this data is highly redundant and can be effectively reduced to a smaller set of variables without losing critical information.

Before applying dimensionality reduction techniques, the raw meteorological data underwent a thorough cleaning and preprocessing phase to ensure accuracy and reliability. The initial dataset, collected monthly, was grouped by climatic seasons (summer, autumn, winter, and spring) to analyze trends and detect inconsistencies such as missing values or noisy data.

To address these issues, the following steps were implemented:

- Detection of Missing Values: The percentage of missing data for each variable was calculated across all seasons, revealing significant gaps in specific time periods that required correction.
- Noise and Outlier Treatment: Outliers were identified using the interquartile range (IQR) method, with thresholds set at a lower bound of 165.0 W/m² and an upper bound of 331.0 W/m². A total of 19,271 outliers were detected, representing 10.16% of the total dataset (189,644 observations). These extreme values were either smoothed through percentile-based techniques, specifically using the 90th percentile threshold, or excluded entirely when their influence on data trends was deemed significant.
- Imputation of Missing Values: For short-term gaps, missing values were imputed using the mean of sur-

rounding observations within each season. For longer-term gaps, controlled noise was added based on a normal distribution, ensuring realistic and non-redundant patterns in the data.

Following this preprocessing, the seasonal data were consolidated into an annual dataset, preserving the integrity of seasonal trends while ensuring a consistent and high-quality foundation for the Principal Component Analysis (PCA) stage. These steps were essential to identify and retain the most representative variables for the subsequent machine learning models. The mathematical procedures that make this variable reduction possible are called dimensionality reduction techniques that are widely developed by Statistics and Machine Learning [26].

In [27] they mentioned that data preprocessing is the central stage in data mining, and dimensionality reduction is a technique used for data preprocessing. Dimensionality reduction involves applying various methods to eliminate excessive and irrelevant features from Machine Learning models. Moreover, it plays a crucial role in identifying and removing features that negatively impact the performance of the Machine Learning model.

Among dimensionality reduction methods, one of the most efficient consists of identifying and selecting a subset of features that are most relevant to the target variable. This approach is known as feature selection, and it is especially effective when dealing with tabular data in which each column represents a specific type of information.

In our case, to reduce and group our meteorological data we used the statistical technique called Principal Component Analysis (PCA). This technique simplifies the analysis, keeping as much relevant information as possible. In [28], it is stated that PCA is a mathematical tool designed to capture and represent the variation within a dataset using a reduced number of factors.

After applying PCA, three components (Component 1, Component 2, and Component 3) were obtained, grouping our variables as follows:

- Component 1 (Thermal sensation): Outside Temperature, Wind Chill, Heat Index, Pressure.
- Component 2 (Humidity): Wind Chill, Dew Point, Relative (Out) Humidity.
- Component 3 (Wind): Wind Dir, Peak Wind Speed.

These three components explain approximately 66.55% of the total variance in the data, with Thermal Sensation accounting for 35.78%, Humidity for 19.41%, and Wind for 11.36%. This highlights their importance in representing the main patterns of variability in the solar radiation dataset.

They constitute the structural model of solar radiation in Lima. Figure 5 shows the structural model obtained from applying the PCA.

According to the PCA criteria, it is determined that our model relies exclusively on the following parameters: Outside Temperature, Wind Chill, Heat Index, Pressure, Dew Point, Relative Humidity, Wind Direction and Peak Wind Speed.

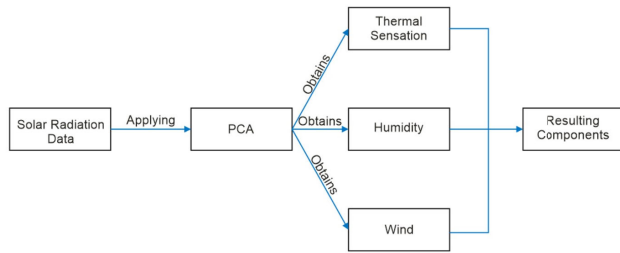


FIGURE 5. Structural model and results of applied analysis.

2) PERFORMANCE METRICS

To validate the performance of the machine learning models applied in this study, a set of commonly used regression metrics was considered: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Coefficient of Determination (R^2), and Mean Absolute Error (MAE). These metrics are widely used to evaluate machine learning models, especially in regression tasks.

In particular, for comparative analysis of models, R^2 is used as it helps to evaluate the explanatory power and the comparison of models between data sets by means of variance. On the other hand, to assess the actual magnitude of the error, RMSE is used for interpretability, or MSE if it is desired to penalize larger errors.

Finally, to avoid sensitivity to outliers, the MAE metric is used, as it provides a balanced view of the mean error.

Additionally, the Mean Absolute Percentage Error (MAPE) was used to assess the relative percentage error between predicted and actual values. MAPE is scale-independent and provides an intuitive interpretation of prediction accuracy. Lower values of MAPE indicate better model performance.

Furthermore, the Nash–Sutcliffe Efficiency (NSE) metric was included to evaluate the predictive power of the models compared to the mean of the observed data. NSE values closer to 1 indicate high predictive ability and better agreement between predictions and observations.

IV. EXPERIMENTAL RESULTS

The dataset comprised 189,644 processed observations were used that were obtained from January to December 2023 between 5 am and 18 pm, which is the period with the highest solar radiation incidence.

The most representative variables used to test our model are the following: OT, Chill, Dew, Heat, Humidity, Pressure, Dir, Pwspeed.

To obtain the results, we apply various Machine Learning algorithms and Ensemble techniques, comparing which model obtains the best metrics in our study.

A. MACHINE LEARNING MODELS FOR SOLAR RADIATION PREDICTION

It is desirable to develop an algorithm capable of predicting solar radiation values based on multiple input variables. The

most appropriate models in our study are the following: Polynomial Regression, Decision Trees, Random Forest, Artificial Neural Networks, K Nearest Neighbors and Gradient Boosting.

1) POLYNOMIAL REGRESSION (PR)

This method extends linear regression by allowing the relationship between the dependent variable and the independent variables to be modeled using higher degree polynomials. It can fit more complex curves, but can become computationally expensive with very high polynomial degrees and can suffer from overfitting.

The MSE and RMSE are relatively high, suggesting that the model may not fit the test data well. The R^2 is moderate, indicating that the model explains approximately 57% of the variability of the data. The MAE is relatively low, suggesting that the model predictions are close to the true values on average, but there may still be discrepancies in some predictions.

Parameters used:

- degree=2. Specifies the degree of the polynomial to use.

Figures 6 and 7 show the analysis of the plot and its validation.

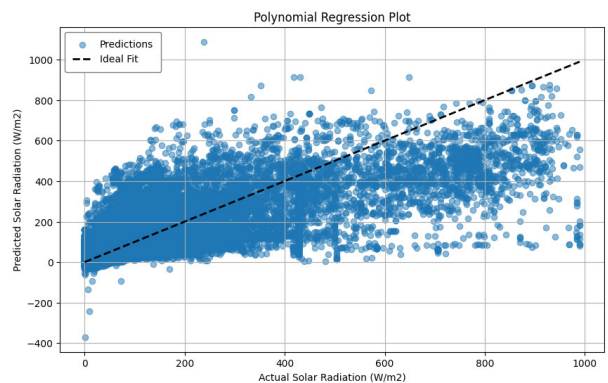


FIGURE 6. Polynomial regression plot.

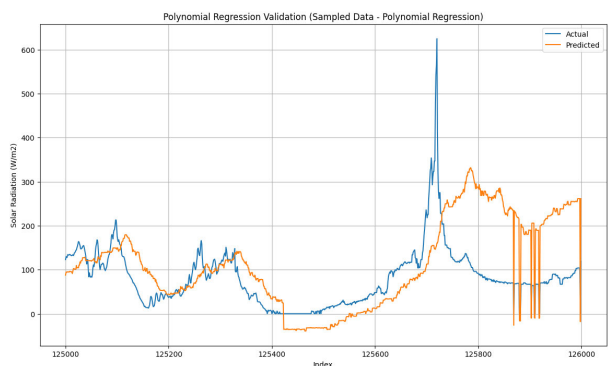


FIGURE 7. Polynomial regression validation.

2) DECISION TREES (DT)

Decision trees are a widely used algorithm for data classification and regression. They are predictive models that use a

tree-like structure to make decisions based on characteristics or attributes of the input data.

It performs noticeably better than polynomial regression with lower MSE and RMSE. An R^2 of 0.87 indicates that the model can explain 87% of the variability in the test data. The MAE of 19.08 shows considerable precision. Decision trees are prone to overfitting, but provide good model interpretation. The execution time obtained is reasonable and fast.

Parameters used:

- `random_state`: 42. This parameter sets a seed for the model's random number generator, ensuring that the results are reproducible.

Figures 8 and 9 show the analysis of the plot and its validation.



FIGURE 8. Decision tree plot.

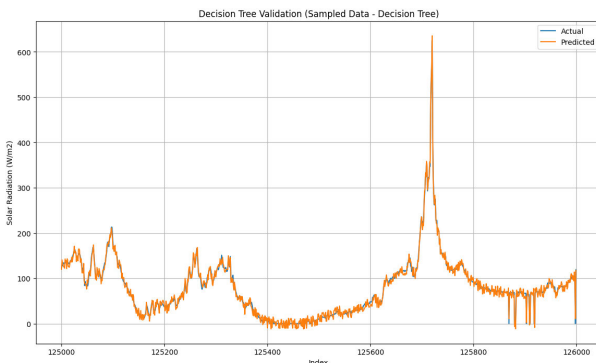


FIGURE 9. Decision tree validation.

3) RANDOM FOREST (RF)

It is a supervised learning algorithm that creates several decision trees using a training dataset. The outcomes from these trees are then combined to produce a single, more robust model than the individual results from each tree [29].

This model demonstrates excellent performance, characterized by very low MSE and RMSE values. An R^2 of 0.91 means that the model can explain 91% of the variability in the test data. The MAE of 18.44 indicates a prediction accuracy similar to that of Decision Trees but with greater

robustness against overfitting. The execution time is longer due to the complexity of training multiple trees, but it is still quite practical.

Parameters used:

- `random_state`: 42. Sets a seed for the random number generator, ensuring that the results are reproducible.

Figures 10 and 11 show the analysis of the plot and its validation.

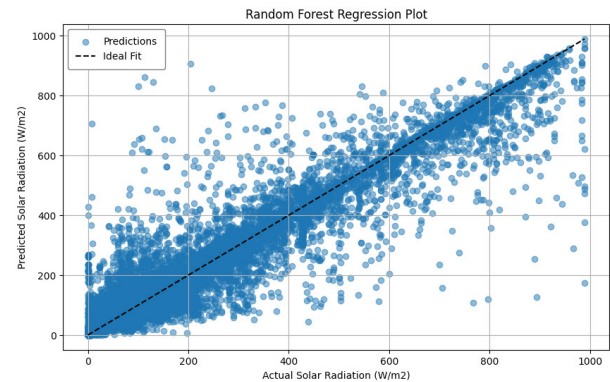


FIGURE 10. Random forest plot.

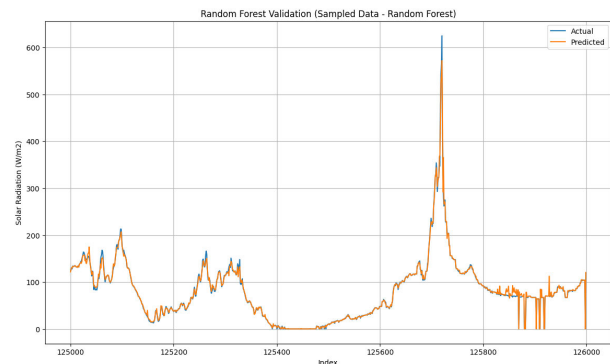


FIGURE 11. Random forest validation.

4) ARTIFICIAL NEURAL NETWORKS (ANN)

Artificial Neural Networks are computational models inspired by the human brain, composed of layers of interconnected neurons. These networks are trained using the error backpropagation algorithm, which adjusts the connection weights between neurons to reduce the error function. Capable of learning non-linear data representations, Artificial Neural Networks are highly effective for various classification and pattern recognition tasks [30].

This model shows moderate performance with relatively high MSE and RMSE values. An R^2 of 0.63 means that the model can explain 63% of the variability in the test data. The MAE of 59.62 is also considerably high. Additionally, the 14-minute runtime is considerably longer compared to other models, which can be a limiting factor in applications where computing time is crucial.

Parameters used:

- layers: Dense: 64 neurons. Dense layers are the basic layers of neural networks.
- activation function: ReLU is a nonlinear activation function that helps the network learn complex nonlinear relationships.
- model.compile: Optimizer 'adam', loss 'mse', metrics 'mae'. The build configures the model for training. Adam is an optimization algorithm that combines the advantages of other algorithms, such as AdaGrad and RMSProp, for efficient updating of model weights.
- model.fit: 100 epochs, batch_size 32, validation_split 0.2. Train the model with the training data. An epoch refers to a complete pass through the entire training dataset. More epochs can lead to better model fit, but also a higher risk of overfitting.

Figures 12 and 13 show the analysis of the plot and its validation.

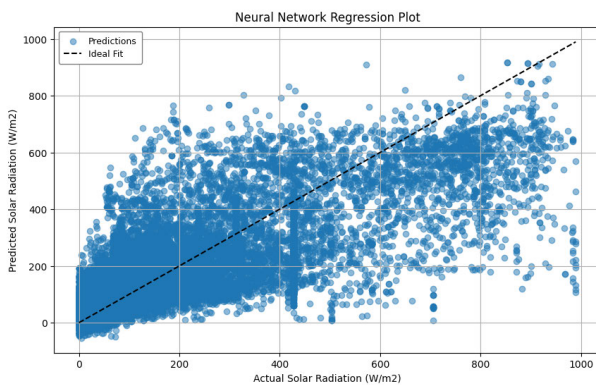


FIGURE 12. Artificial neural networks plot.

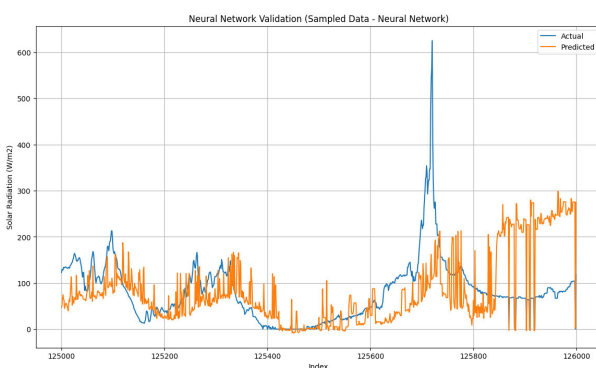


FIGURE 13. Artificial neural networks validation.

5) K NEAREST NEIGHBORS (KNN)

It is an algorithm within supervised learning that primarily functions to classify values by identifying the “most similar” (closest) data points learned during the training phase and making predictions for new points based on that classification.

The KNN model performs well with higher MSE and RMSE than Random Forest and Decision Trees. An R^2 of 0.75 means that the model can explain 75% of the variability in the test data. The MAE of 41.91 shows moderate accuracy. The KNN is quite fast to run, but its performance can decrease with large data sets.

Parameters used:

- n_neighbors: 5. Number of neighbors to use for prediction, 5 is the default value.
- weights: 'uniform'. Weight function for prediction. 'uniform' means that all neighbors have the same influence.
- algorithm: 'auto'. Algorithm used to calculate nearest neighbors. 'auto' selects the most efficient method based on the value of n_neighbors and the characteristics of the data set.
- leaf_size: 30. Size of the sheet passed to the BallTree or KDTree. Affects the speed of tree construction and query.
- p:2. Power of the Minkowski parameter. When $p = 2$, the Euclidean distance is used.

Figures 14 and 15 show the analysis of the plot and its validation.

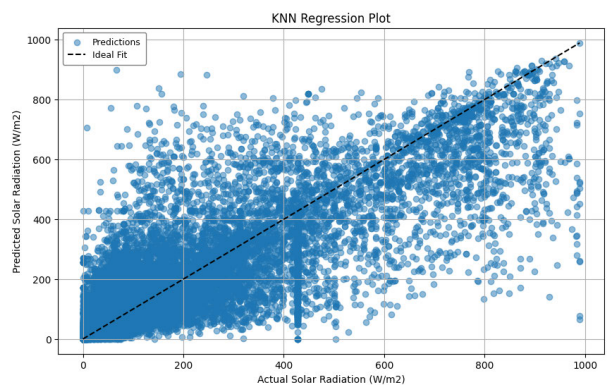


FIGURE 14. K Nearest neighbors plot.

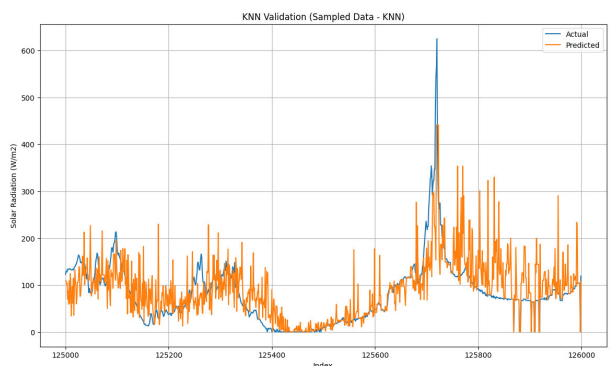


FIGURE 15. K-Nearest neighbors validation.

6) GRADIENT BOOSTING (GB)

It is a supervised learning algorithm that builds a more robust model from the combination of multiple simpler decision

trees. In each iteration, the model is adjusted and the residual errors of the previous model decrease. The term gradients is related to the optimization technique used to minimize the loss function [31].

This model achieved moderate performance, with an R^2 of 0.64, indicating it explains 64% of the variability in the test data. The MSE and RMSE are relatively high, indicating that the predictions are not very accurate. The execution time is reasonable, making Gradient Boosting a viable option when seeking a balance between accuracy and time efficiency. Parameters used:

- random_state: 42. Fixes the seed to ensure reproducibility of results.
- loss: 'ls'. Loss function to optimize 'ls' (least squares) for regression.
- learning_rate: 0.1. Learning rate that reduces the contribution of each tree. A lower value may lead to better results, but requires more iterations.
- n_estimators: 100. Number of decision trees to build. A higher value may improve accuracy but increases training time.
- subsample: 1.0. Fraction of samples used to train each tree. Subsampling can reduce variance and avoid overfitting.
- criterion: 'friedman_mse'. Criterion to measure the quality of division in trees, 'friedman_mse' is the default.
- min_samples_split: 2. Minimum number of samples required to split a node.
- min_samples_leaf: 1. Minimum number of samples needed on one sheet.
- max_depth: 3. Maximum depth of trees. Limiting depth can help avoid overfitting.

Figures 16 and 17 show the analysis of the plot and its validation.

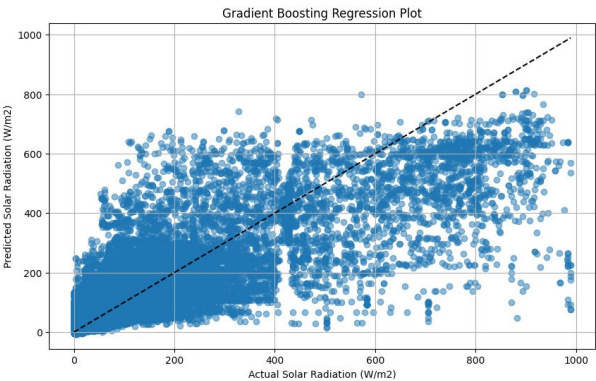


FIGURE 16. Gradient boosting plot.

All these results are presented in Table 3. From Table 3, it can be observed that among all evaluated models, the Random Forest (RF) algorithm yielded the best performance for predicting meteorological variables during 2023 in the Rimac district, Lima-Peru. Its ability to model

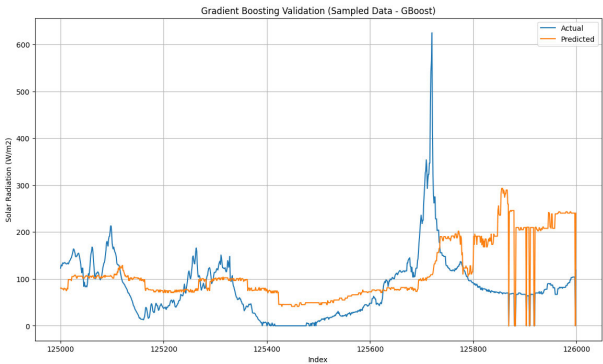


FIGURE 17. Gradient boosting validation.

TABLE 3. Annual results of solar radiation prediction for 2023.

Model	MSE	RMSE	R^2	MAE	NSE	MAPE	Time (s)
PR	12558.69	112.07	0.57	67.95	0.5646	57.91%	2.86
DT	3807.90	61.71	0.87	19.08	0.8688	31.95%	1.49
RF	2241.09	47.34	0.91	18.44	0.9122	32.35%	77.61
ANN	10574.83	102.83	0.63	59.62	0.6269	25.22%	840.00
KNN	7295.00	85.41	0.75	41.91	0.7471	72.64%	2.97
GB	10368.01	101.82	0.65	60.67	0.6405	34.60%	26.47

non-linear relationships and mitigate overfitting makes it particularly effective across various metrics, achieving the lowest MSE and RMSE, the highest values for R^2 and NSE, and a MAPE of 32.35%, which is generally considered to represent reasonable prediction accuracy.

Although RF has a higher execution time compared to simpler models, its precision and robustness justify its use in accuracy-critical applications.

Other models, such as Decision Tree (DT) and K-Nearest Neighbors (KNN), also showed good predictive performance. DT achieved competitive accuracy with a relatively low MAPE (31.95%) and the shortest computation time, making it suitable for scenarios where speed is essential. KNN, on the other hand, exhibited the highest MAPE (72.64%), indicating limited reliability despite a short execution time.

Gradient Boosting (GB) and Artificial Neural Networks (ANN) demonstrated moderate performance. While capable of capturing complex patterns, their MAPE values were above 25%, and their higher computational cost did not yield significantly better predictive accuracy compared to RF.

Each model presents specific strengths and limitations. Therefore, the choice of the most suitable model should consider the application's particular needs, including required accuracy, interpretability, and computational constraints. For real-time or large-scale applications, faster models like KNN or GB may be preferable, while for tasks demanding higher accuracy and robustness, Random Forest stands out as the optimal choice.

B. ENSEMBLE TECHNIQUES

In [32] mentioned that one of the main problems in ML is generalization. Generalization refers to the predictive ability of the ML. The authors mentioned that the better a

predictor performs with the data, the greater has the ability to generalize. In Machine Learning, an Ensemble is a method that merges several models to create a stronger model with improved performance compared to any single model on its own.

In [33] mentioned that it has been formally and empirically shown that ensembles of learning machines generalize better to individual predictors in many cases. Independently in [34] mentioned that Ensemble techniques reduce the bias and variance of our model.

Several ensemble techniques exist, including Voting, Stacking, Bagging [35] and Boosting [36]. We use some of these techniques and apply them to various combinations of different algorithms to achieve the best results. In our case we operate with the following models: Voting Regressor (VR), Stacking Regressor (SR), Bagging (Ba). The corresponding results are summarized in Table 4.

TABLE 4. Various models of ensembles.

Model	MSE	RMSE	R ²	MAE	NSE	MAPE	Time (s)
VR(RF,DT,KNN,GB)	3605.35	60.04	0.88	31.88	0.8751	63.93%	99.35
VR(RF,KNN,GB)	4597.51	67.80	0.84	37.67	0.8406	76.51%	100.53
VR(RF,DT)	2622.44	51.21	0.91	18.02	0.9092	31.13%	83.29
VR(RF,KNN)	3559.04	59.66	0.88	28.76	0.8766	51.00%	84.06
VR(RF,GB)	4352.14	65.97	0.85	37.94	0.8491	81.44%	96.62
VR(KNN,GB)	7409.25	86.08	0.74	48.85	0.7431	52.00%	25.26
SR(RF,DT,KNN,GB)	2238.60	47.30	0.92	18.27	0.9242	32.03%	508.09
SR(RF,KNN,GB)	2238.66	47.31	0.92	18.28	0.9240	32.04%	500.58
SR(DT,KNN,GB)	3342.44	57.81	0.88	24.06	0.8764	43.20%	143.20
SR(RF,KNN)	2240.28	47.33	0.92	18.34	0.9231	32.13%	372.22
SR(RF,GB)	2239.00	47.32	0.92	18.35	0.9222	32.30%	484.36
SR(KNN,GB)	6989.66	83.60	0.76	44.32	0.7577	81.31%	141.62
SR(RF,DT)	2241.88	47.35	0.92	18.51	0.9221	32.81%	365.70
BR(RF)	2461.01	49.61	0.91	21.67	0.9144	38.69%	180.00
BR(DT)	2471.56	49.71	0.91	19.29	0.9130	33.76%	180.00
VBR(RF, DT)	2373.79	48.72	0.92	20.13	0.9178	35.76%	540.00

From the results, it is observed that ensemble models using the Stacking Regressor (SR) with three or four base algorithms achieve the best performance, with lower MSE, RMSE, and MAE, and higher R^2 , NSE, and lower MAPE values—indicating enhanced accuracy and generalization capability.

For instance, the SR(RF, DT, KNN, GB) model achieved an NSE of 0.9242 and a MAPE of 32.03%, which are commonly interpreted as indicating very good predictive performance and reasonable accuracy, respectively. This supports the effectiveness of ensemble techniques in improving model robustness.

Voting Regressor (VR) and Bagging Regressor (BR) models—such as VR(RF, DT) or BR(RF)—also outperform individual models from Table 3, though their metrics are generally slightly less favorable than those of SR-based models.

However, these performance gains come at a computational cost. SR-based ensembles, while more accurate, require significantly more execution time, as seen in the last column of Table 4.

Therefore, selecting an ensemble method should balance prediction quality with computational efficiency. For applications requiring precision, SR(RF, DT, KNN, GB) is recommended. For time-sensitive or limited-resource scenarios, simpler combinations like VR(RF, DT) may be more practical.

Figures 18 and 19 shows diagrams of some Ensembles models.

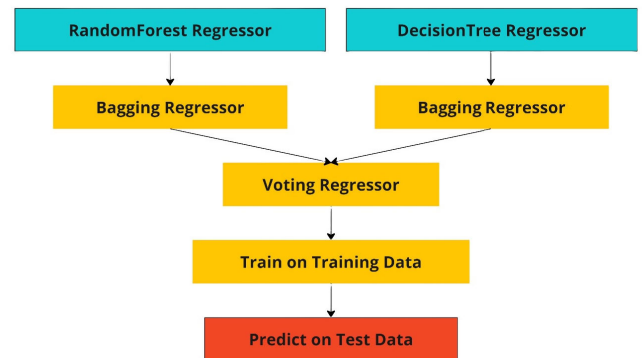


FIGURE 18. Voting regressor.

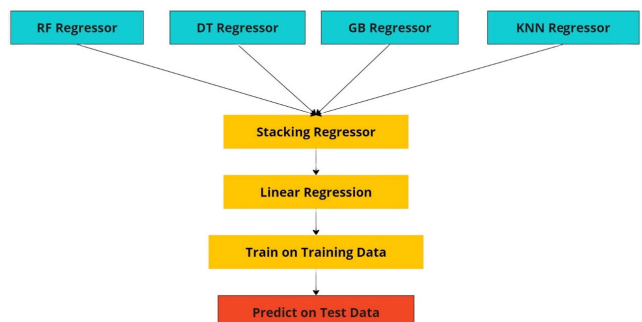


FIGURE 19. Stacking regressor.

Figures 20 and 21 show the analysis of the plot and its validation of the chosen Stacking Regressor model.

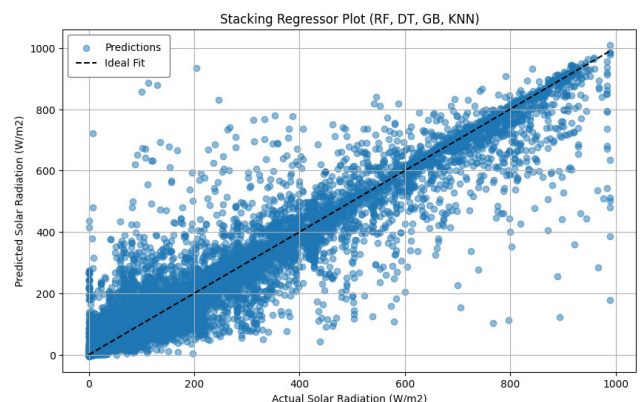


FIGURE 20. Stacking regressor plot.

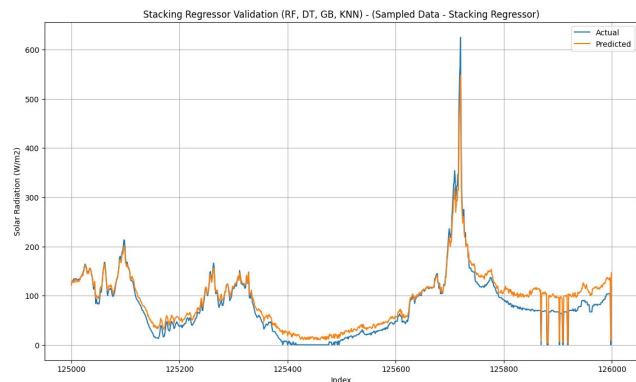


FIGURE 21. Stacking regressor validation.

Then we present part of the Stacking Regressor code developed. This code presents the implementation of an ensemble regression model using the Stacking technique.

In this approach, multiple base models (Random Forest, Decision Tree, Gradient Boosting, and K-Nearest Neighbors) are combined, and their predictions are integrated through a Linear Regression meta-model.

This strategy aims to enhance predictive performance by leveraging the individual strengths of each estimator.

```
# Importing libraries
from sklearn.ensemble import StackingRegressor
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import GradientBoostingRegressor, RandomForestRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.neighbors import KNeighborsRegressor

estimators = [
    ('rf', RandomForestRegressor(random_state=42)),
    ('dt', DecisionTreeRegressor(random_state=42)),
    ('gb', GradientBoostingRegressor(random_state=42)),
    ('knn', KNeighborsRegressor())
]

final_estimator = LinearRegression()
stacking_regressor = StackingRegressor(
    estimators=estimators,
    final_estimator=final_estimator
)

stacking_regressor.fit(X_train_mod_scaled,
    y_train_mod)
```

LISTING 1. Stacking regressor implementation.

This code shows the performance evaluation of the Stacking Regressor on the test dataset using common regression metrics, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Coefficient of Determination (R^2), and Mean Absolute Error (MAE). These metrics provide

a comprehensive understanding of the model's prediction accuracy.

```
# Evaluate the model
y_pred_stacking =
    stacking_regressor.predict(X_test_mod_scaled)
mse_stacking = mean_squared_error(y_test_mod,
    y_pred_stacking)
r2_stacking = r2_score(y_test_mod,
    y_pred_stacking)
mae_stacking = mean_absolute_error(y_test_mod,
    y_pred_stacking)
rmse_stacking = np.sqrt(mse_stacking)

print(f"MSE: {mse_stacking}, RMSE:
    {rmse_stacking}, "
    f"R2: {r2_stacking}, MAE: {mae_stacking}")
```

LISTING 2. Evaluation of the stacking regressor on the test set.

The selected ensemble model, leveraging Random Forest, Decision Tree, K-Nearest Neighbors, and Gradient Boosting, demonstrated exceptional performance with an R^2 of 0.92 on the test set. Additionally, the cross-validation results further validated the model's robustness and generalization. These findings highlight the model's potential for real-world applications, particularly in optimizing solar energy systems in regions with climatic conditions similar to those of Lima. This is shown in Figure 22.

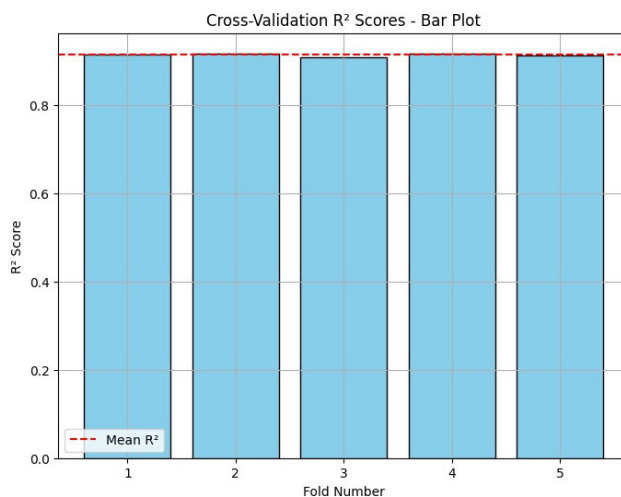


FIGURE 22. Cross-validation R^2 scores across folds.

Furthermore, to analyze how the model's performance scales with data size, a learning curve was generated. Figure 23 demonstrates that both training and validation R^2 scores converge as the training set size increases, confirming the model's ability to leverage larger datasets without overfitting.

V. DISCUSSION OF RESULTS

Several previous studies have addressed solar radiation prediction using machine learning models, as summarized in

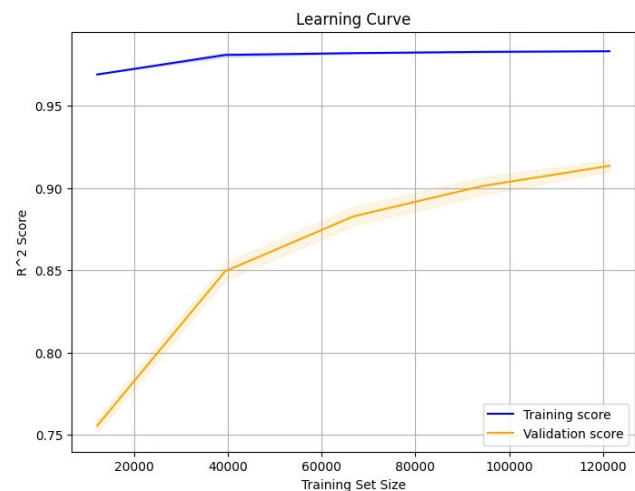


FIGURE 23. Learning curve for stacking regressor: Training vs Validation scores.

Table 5. This table compares various approaches, including Artificial Neural Networks (ANN), Support Vector Regression (SVR), Random Forest (RF), NARX, and LSTM, achieving different R^2 values depending on the set of input variables used. However, many of these studies were conducted in different climate contexts, such as tropical climates or regions with greater access to cloud cover and duration data of the sunshine.

TABLE 5. Comparison of works predicting solar radiation.

Author	Model	Variables	R^2
Citakoglu [37]	ANN	Month, Temp, RH, Extraterrestrial radiation	94%
Sreekumar [38]	SVR	Cloud Cover, Cloud Move., Wind Spd., Temp.	97.11%
Rahman [39]	ANN	Temp., Press., Relative Humidity	98%
Fuselero [40]	NARX	Avg Temp., Rainfall, RH, Wind Dir., Wind Spd., Sunshine Dur.	98%
Villegas-Mier [41]	RF	Temp., Press., RH, Sunshine Hrs	95.98%
Chaibi [43]	RF	Temp., Press., RH, Rain, Wind Spd., Sunshine Hrs	94%
Zuo [44]	LSTM	Press., Wind Spd., Temp., RH, AOD	90%
Segovia [45]	RF	Temp., RH, Wind Spd.	73.33%
Solano [42]	Voting	Wind Spd., Wind Dir., Press., Dew point, RH	84.8%
Proposed model	SR	Temp., Wind Chill, Heat index, Press., Dew point, RH, Wind Dir, Peak Wind Spd	92.22%

In contrast, this study focuses on a subtropical desert climate, specifically in the district of Rímac, Lima, a region characterized by low rainfall, high humidity, and persistent cloudiness—conditions that are not widely represented in the solar radiation prediction literature. Despite working with a limited set of locally available variables, the proposed model, based on Ensemble Learning (Stacking Regressor),

achieved an R^2 of 0.92, RMSE of 47.30 W/m², and MAE of 18.27 W/m², placing it at a competitive level compared to the reference models in Table 5.

Principal Component Analysis (PCA) was fundamental for dimensionality reduction, allowing the selection of the 8 most representative variables from the original 11. This technique grouped the variables into three main components: thermal sensation, humidity, and wind, optimizing both the performance and the interpretability of the model. However, the absence of variables such as cloud cover or sunshine duration, which are relevant for estimating solar radiation, reflects the limitations of data availability at the selected meteorological station.

In addition, the proposed approach features a less complex computational architecture, making it suitable for implementation in low-resource environments, such as rural communities without access to advanced meteorological stations. This is a crucial aspect, as many ensemble models like LSTM and ANN require substantial computational resources, limiting their applicability in areas with less technological infrastructure.

Despite the good results obtained, it is important to note that the model was validated using data from only one year (2023), which may limit its long-term generalization. Models trained on longer time series are more effective in capturing seasonal variability and extreme weather events. Therefore, future research should include multi-year datasets to better assess the model’s ability to generalize over time and across different conditions.

A. ALIGNMENT WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs)

This research aligns with “Sustainable Development Goal 13 (SDG 13)”, established by the United Nations in 2015, which aims to “Take urgent action to combat climate change and its impacts.” Solar energy is a key component in reducing reliance on fossil fuels, thereby mitigating greenhouse gas emissions and supporting climate resilience in vulnerable regions. Specifically, this work contributes to several of SDG 13’s targets:

- **Strengthening resilience:** By providing accurate solar radiation predictions in arid, low-rainfall areas, this study enables informed planning of solar energy systems that increase regional resilience to climate-related challenges.
- **Integrating climate change measures into planning:** The methodology presented can support local governments and energy planners in designing renewable energy solutions tailored to their region’s meteorological conditions.
- **Building capacity and knowledge:** The public availability of data and source code improves knowledge sharing and capacity building for institutions aiming to implement similar systems.
- **Promoting mechanisms for planning and management:** This study exemplifies how data-driven approaches,

powered by machine learning, can enhance climate-sensitive decision-making.

Overall, the predictive models developed in this study help to overcome limitations in data collection infrastructure in isolated areas, making solar energy projects more feasible and aligned with global sustainability efforts [6].

B. LIMITATIONS AND RECOMMENDATIONS FOR FUTURE WORK

This study presents several limitations that may affect the generalizability, scalability, and broader applicability of the proposed model.

First, the model was trained and validated using data from a single year (2023), which may limit its capacity to capture seasonal variability and interannual trends. As such, its performance in different meteorological years remains to be tested. Additionally, preliminary tests using early 2024 data indicate that the model retains acceptable performance when applied to new seasonal conditions. A more extensive temporal and spatial validation will be addressed in future work.

Second, while the Principal Component Analysis (PCA) allowed for dimensionality reduction, some relevant meteorological variables—such as cloud cover, sunshine duration, and diffuse radiation—were not available from the local weather station and thus were excluded. Their inclusion could potentially enhance model accuracy and responsiveness to atmospheric dynamics.

Third, computational demands may limit model implementation in low-resource environments. Although the models were executed on a high-performance university workstation, ensemble techniques such as Stacking Regressor are resource-intensive, which may hinder its deployment in low-resource or real-time environments.

Fourth, the model has only been validated in a subtropical desert climate (Lima, Peru). Its applicability in other climatic regions with distinct meteorological behaviors (e.g., tropical, temperate, continental) remains untested.

To address these limitations, we propose the following recommendations for future work:

- Incorporate multi-year datasets to improve temporal generalization.
- Expand the input variable set to include cloud-related and solar irradiance-specific variables.
- Develop lightweight versions of the model for deployment on low-power hardware in rural areas.
- Validate the model in diverse geographic and climatic regions to assess its robustness and transferability.

VI. CONCLUSION

This study proposed an efficient machine learning-based approach for solar radiation prediction in arid urban environments, using Lima, Peru as a case study. The methodology integrated key meteorological variables and employed dimensionality reduction through Principal

Component Analysis (PCA) to enhance model simplicity and interpretability.

Among the tested models, the Stacking Regressor yielded the best performance, confirming the potential of ensemble techniques in capturing nonlinear relationships among weather parameters. The results highlight the feasibility of accurate solar radiation forecasting even in regions with limited data availability and measurement infrastructure.

The approach is particularly relevant for remote and low-resource areas where traditional weather monitoring systems are scarce. Moreover, the study contributes to the goals of sustainable development, particularly SDG 13 (Climate Action), by supporting decision-making in solar energy planning and promoting the adoption of renewable energy technologies.

Future work should focus on incorporating multi-year datasets, expanding the range of input variables (e.g., cloud cover, sunshine duration), and testing the model in diverse climatic regions to further assess its generalizability and robustness. Preliminary tests using early 2024 data suggest that the model maintains acceptable performance when applied to new seasonal conditions. These findings will be further explored in upcoming research.

These results confirm the high reliability of the Ensemble (RF, DT, KNN, GB) prediction model, particularly in subtropical desert climates like Lima, Peru.

REFERENCES

- [1] J.-I. Prieto and D. García, "Global solar radiation models: A critical review from the point of view of homogeneity and case study," *Renew. Sustain. Energy Rev.*, vol. 155, Mar. 2022, Art. no. 111856.
- [2] C. Zhang, L. Hua, C. Ji, M. S. Nazir, and T. Peng, "An evolutionary robust solar radiation prediction model based on WT-CEEMDAN and IASO-optimized outlier robust extreme learning machine," *Appl. Energy*, vol. 322, Sep. 2022, Art. no. 119518.
- [3] P. Chaudhary, R. Gattu, S. Ezekiel, and J. A. Rodger, "Forecasting solar radiation: Using machine learning algorithms," *J. Cases Inf. Technol. (JCIT)*, vol. 23, no. 4, pp. 1–21, Jan. 2021.
- [4] M. K. Nematchoua, J. A. Orosa, and M. Afaifia, "Prediction of daily global solar radiation and air temperature using six machine learning algorithms; a case of 27 European countries," *Ecol. Informat.*, vol. 69, Jul. 2022, Art. no. 101643.
- [5] A. Geetha, J. Santhakumar, K. M. Sundaram, S. Usha, T. M. T. Thentral, C. S. Boopathi, R. Ramya, and R. Sathyamurthy, "Prediction of hourly solar radiation in Tamil Nadu using ANN model with different learning algorithms," *Energy Rep.*, vol. 8, pp. 664–671, Apr. 2022.
- [6] *The Sustainable Development Goals Report 2020*, United Nations, New York, NY, USA, 2020.
- [7] D. Cui, S. Liang, and D. Wang, "Observed and projected changes in global climate zones based on Köppen climate classification," *WIREs Climate Change*, vol. 12, no. 3, p. e701, May 2021.
- [8] M. Kottek, J. Grieser, C. Beck, B. Rudolf, and F. Rubel, "World map of the Köppen–Geiger climate classification updated," *Meteorologische Zeitschrift*, vol. 15, no. 3, pp. 259–263, Jul. 2006.
- [9] W. Köppen, "Versuch einer Klassifikation der Klimate, vorzugsweise nach ihren Beziehungen zur Pflanzenwelt," *Geographische Zeitschrift*, vol. 6, no. 11, pp. 593–611, 1900.
- [10] H. Verbois, R. Huva, A. Rusydi, and W. Walsh, "Solar irradiance forecasting in the tropics using numerical weather prediction and statistical learning," *Sol. Energy*, vol. 162, pp. 265–277, Mar. 2018.

- [11] M. Husein and I.-Y. Chung, "Day-ahead solar irradiance forecasting for microgrids using a long short-term memory recurrent neural network: A deep learning approach," *Energies*, vol. 12, no. 10, p. 1856, May 2019.
- [12] O. Goncu, T. Koroglu, and N. F. Ozdil, "Estimation of hourly global solar radiation using artificial neural network in Adana Province, Turkey," *J. Thermal Eng.*, vol. 7, no. Supp 14, pp. 2017–2030, Dec. 2021.
- [13] A. Rabeih, M. Guermoui, and D. Lalmi, "Hybrid models for global solar radiation prediction: A case study," *Int. J. Ambient Energy*, vol. 41, no. 1, pp. 31–40, Jan. 2020.
- [14] Z. Pang, F. Niu, and Z. O'Neill, "Solar radiation prediction using recurrent neural network and artificial neural network: A case study with comparisons," *Renew. Energy*, vol. 156, pp. 279–289, Aug. 2020.
- [15] D. Guijo-Rubio, A. M. Durán-Rosal, P. A. Gutiérrez, A. M. Gómez-Orellana, C. Casanova-Mateo, J. Sanz-Justo, S. Salcedo-Sanz, and C. Hervás-Martínez, "Evolutionary artificial neural networks for accurate solar radiation prediction," *Energy*, vol. 210, Nov. 2020, Art. no. 118374.
- [16] H. Citakoglu, B. Babayigit, and N. A. Haktanir, "Solar radiation prediction using multi-gene genetic programming approach," *Theor. Appl. Climatol.*, vol. 142, nos. 3–4, pp. 885–897, Nov. 2020.
- [17] Ü. Ağbulut, A. E. Gürel, and Y. Biçen, "Prediction of daily global solar radiation using different machine learning algorithms: Evaluation and comparison," *Renew. Sustain. Energy Rev.*, vol. 135, Jan. 2021, Art. no. 110114.
- [18] A. Kuhe, V. T. Achirgenda, and M. Agada, "Global solar radiation prediction for Makurdi, Nigeria, using neural networks ensemble," *Energy Sour., A, Recovery, Utilization, Environ. Effects*, vol. 43, no. 11, pp. 1373–1385, Jun. 2021.
- [19] R. Elmousaid, Y. Adnani, A. El Hamdaoui, and R. Elgouri, "Daily solar radiation prediction using NARX and MLP-NNs networks: A case study of Kenitra city, Morocco," in *Proc. IEEE 3rd Int. Conf. Electron., Control, Optim. Comput. Sci. (ICECOCS)*, Dec. 2022, pp. 1–6.
- [20] S. Ghimire, R. C. Deo, D. Casillas-Pérez, and S. Salcedo-Sanz, "Boosting solar radiation predictions with global climate models, observational predictors and hybrid deep-machine learning algorithms," *Appl. Energy*, vol. 316, Jun. 2022, Art. no. 119063.
- [21] M. M. Comert, K. Adem, and M. Erdogan, "Comparative analysis of estimated solar radiation with different learning methods and empirical models," *Atmósfera*, vol. 37, pp. 273–284, Aug. 2023.
- [22] V. Demir and H. Citakoglu, "Forecasting of solar radiation using different machine learning approaches," *Neural Comput. Appl.*, vol. 35, no. 1, pp. 887–906, Jan. 2023.
- [23] A. Dahmani, Y. Ammi, S. Hanini, M. R. Yaiche, and H. Zentou, "Prediction of hourly global solar radiation: Comparison of neural networks/bootstrap aggregating," *Kemija u industriji, Časopis za kemijsku industriju*, vol. 72, nos. 3–4, p. 201, Mar. 2023.
- [24] F. Gerges, M. C. Boufadel, E. Bou-Zeid, H. Nassif, and J. T. L. Wang, "Long-term prediction of daily solar irradiance using Bayesian deep learning and climate simulation data," *Knowl. Inf. Syst.*, vol. 66, no. 1, pp. 613–633, Jan. 2024.
- [25] F. D. Alay, N. Ilhan, and M. T. Güllüoğlu, "A comparative study of data mining methods for solar radiation and temperature forecasting models," *JUCS-J. Universal Comput. Sci.*, vol. 30, no. 6, pp. 847–877, Jun. 2024.
- [26] C. O. S. Sorzano, J. Vargas, and A. Pascual Montano, "A survey of dimensionality reduction techniques," 2014, *arXiv:1403.2877*.
- [27] H. S. Obaid, S. A. Dheyab, and S. S. Sabry, "The impact of data pre-processing techniques and dimensionality reduction on the accuracy of machine learning," in *Proc. 9th Annu. Inf. Technol., Electromechanical Eng. Microelectron. Conf. (IEMECON)*, Mar. 2019, pp. 279–283.
- [28] D. Granato, J. S. Santos, G. B. Escher, B. L. Ferreira, and R. M. Maggio, "Use of principal component analysis (PCA) and hierarchical cluster analysis (HCA) for multivariate association between bioactive compounds and functional properties in foods: A critical perspective," *Trends Food Sci. Technol.*, vol. 72, pp. 83–90, Feb. 2018.
- [29] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, pp. 5–32, Jan. 2001.
- [30] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.
- [31] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, Oct. 2001.
- [32] A. Chandra, "Evolutionary approach to tackling the trade-off between diversity and accuracy in neural network ensembles," *School Comput. Sci., Univ. Birmingham, U.K., Tech. Rep.*, 2004, vol. 431.
- [33] G. Brown, J. Wyatt, R. Harris, and X. Yao, "Diversity creation methods: A survey and categorisation," *Inf. Fusion*, vol. 6, no. 1, pp. 5–20, Mar. 2005.
- [34] G. Valentini and F. Masulli, "Ensembles of learning machines," in *Proc. 13th Italian Workshop Neural Nets*, Vietri sul Mare, Italy. Berlin, Germany: Springer, Jun. 2002, pp. 3–20.
- [35] L. Breiman, "Bagging predictors," *Mach. Learn.*, vol. 24, no. 2, pp. 123–140, Aug. 1996.
- [36] Y. Freund and R. E. Schapire, "A short introduction to boosting," *J.-Jpn. Soc. For Artif. Intell.*, vol. 14, nos. 771–780, p. 1612, Jan. 1999.
- [37] H. Citakoglu, "Comparison of artificial intelligence techniques via empirical equations for prediction of solar radiation," *Comput. Electron. Agricult.*, vol. 118, pp. 28–37, Oct. 2015.
- [38] S. Sree Kumar, K. C. Sharma, and R. Bhakar, "Optimized support vector regression models for short term solar radiation forecasting in smart environment," in *Proc. IEEE Region 10 Conf. (TENCON)*, Nov. 2016, pp. 1929–1932.
- [39] S. Rahman, S. Rahman, and A. K. M. B. Haque, "Prediction of solar radiation using artificial neural network," *J. Phys., Conf. Ser.*, vol. 1767, no. 1, Feb. 2021, Art. no. 012041.
- [40] A. D. P. Fuselero, H. M. S. A. Portus, and B. T. Doma Jr., "Daily solar radiation forecasting based on a hybrid NARX-GRU network in Dumaguete, Philippines," *Int. J. Renew. Energy Develop.*, vol. 11, no. 3, pp. 839–850, Aug. 2022.
- [41] C. G. Villegas-Mier, J. Rodríguez-Resendiz, J. M. Álvarez-Alvarado, H. Jiménez-Hernández, and Á. Odry, "Optimized random forest for solar radiation prediction using sunshine hours," *Micromachines*, vol. 13, no. 9, p. 1406, Aug. 2022.
- [42] E. S. Solano, P. Dehghanian, and C. M. Affonso, "Solar radiation forecasting using machine learning and ensemble feature selection," *Energies*, vol. 15, no. 19, p. 7049, Sep. 2022.
- [43] M. Chaibi, E. M. Benghoulam, L. Tarik, M. Berrada, and A. El Hmaidi, "Machine learning models based on random forest feature selection and Bayesian optimization for predicting daily global solar radiation," *Int. J. Renew. Energy Develop.*, vol. 11, no. 1, pp. 309–323, Feb. 2022.
- [44] H.-M. Zuo, J. Qiu, Y.-H. Jia, Q. Wang, and F.-F. Li, "Ten-minute prediction of solar irradiance based on cloud detection and a long short-term memory (LSTM) model," *Energy Rep.*, vol. 8, pp. 5146–5157, Nov. 2022.
- [45] J. A. Segovia, J. F. Toaquiza, J. R. Llanos, and D. R. Rivas, "Meteorological variables forecasting system using machine learning and open-source software," *Electronics*, vol. 12, no. 4, p. 1007, Feb. 2023.



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